

ASSOCIATION ANALYSIS

EVALUATION – PART III

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EVALUATION

The following concepts will be introduced:

- ✓ **APPARENT INCONSISTENCY OF ASSOCIATION ANALYSIS**

EVALUATION: BEYOND PAIRS OF BINARY ATTRIBUTES 1

You ask to the Director of the mini-market whether it is relevant to **generate association rules including more than one antecedent**.



The **DIRECTOR OF THE MINI-MARKET** answers that in the past he was puzzled by a **curious finding** of a consultancy company he payed for helping to make a decision about products positioning.

The **curious finding** originates from considering the following items; **beer** and **hot dogs**

The **CONSULTANCY COMPANY** found that

customers who buy beer are less likely to buy hot dogs than those who do not buy beer

However, the **DIRECTOR OF THE MINI-MARKET** knows from his working experience, (**another mini-market**)

customers who buy beer are more likely to buy hot dogs than those who do not buy beer



EVALUATION: BEYOND PAIRS OF BINARY ATTRIBUTES 2



You use the transaction data and compute the following tables

		hot dogs	NOT hot dogs	Total
single	beer	30	10	40
	NOT beer	5	40	45
NOT single	beer	40	88	128
	NOT beer	40	15	55

		hot dogs	NOT hot dogs	Total
beer	hot dogs	70	98	168
	NOT beer	45	55	100
		115	153	



You use the transaction data and compute the following tables

	hot dogs	NOT hot dogs	Total
beer	70	98	168
NOT beer	45	55	100
	115	153	

The right table allows to extract the following rules

{beer} → {hot dogs}

with confidence 42%



You use the transaction data and compute the following tables

	hot dogs	NOT hot dogs	Total
beer	70	98	168
NOT beer	45	55	100
	115	153	

The right table allows to extract the following rules

{beer} → {hot dogs}

with confidence 42% **(70/168)**

EVALUATION: BEYOND PAIRS OF BINARY ATTRIBUTES 2



You use the transaction data and compute the following tables

	hot dogs	NOT hot dogs	Total
beer	70	98	168
NOT beer	45	55	100
	115	153	

The right table allows to extract the following rules

{beer} → {hot dogs}

with confidence 42%

{NOT beer} → {hot dogs}

with confidence 45%

EVALUATION: BEYOND PAIRS OF BINARY ATTRIBUTES 2



You use the transaction data and compute the following tables

	hot dogs	NOT hot dogs	Total
beer	70	98	168
NOT beer	45	55	100
	115	153	

The right table allows to extract the following rules

$\{\text{beer}\} \rightarrow \{\text{hot dogs}\}$

with confidence 42%

$\{\text{NOT beer}\} \rightarrow \{\text{hot dogs}\}$

with confidence 45% **(45/100)**

EVALUATION: BEYOND PAIRS OF BINARY ATTRIBUTES 2



You use the transaction data and compute the following tables

	hot dogs	NOT hot dogs	Total
beer	70	98	168
NOT beer	45	55	100
	115	153	

The right table allows to extract the following rules

{beer} → **{hot dogs}**

with confidence **42%**

{NOT beer} → **{hot dogs}**

with confidence **45%**



Customers who buy beer **{beer}** are less likely (**42%**) to buy hot dogs **{hot dogs}** than those who do not buy beer **{NOT beer}** (**45%**)



You use the transaction data and compute the following tables

		hot dogs	NOT hot dogs	Total
single	beer	30	10	40
	NOT beer	5	40	45
NOT single	beer	40	88	128
	NOT beer	40	15	55

The left table allows to extract the following pairs of rules



{beer} → {hot dogs} with confidence 75%
{NOT beer} → {hot dogs} with confidence 11%



You use the transaction data and compute the following tables

		hot dogs	NOT hot dogs	Total
single	beer	30	10	40
	NOT beer	5	40	45
NOT single	beer	40	88	128
	NOT beer	40	15	55

The left table allows to extract the following pairs of rules



{beer} → {hot dogs} with confidence 75% **(30/40)**
{NOT beer} → {hot dogs} with confidence 11% **(5/45)**



You use the transaction data and compute the following tables

		hot dogs	NOT hot dogs	Total
single	beer	30	10	40
	NOT beer	5	40	45
NOT single	beer	40	88	128
	NOT beer	40	15	55

The left table allows to extract the following pairs of rules

{beer} → {hot dogs} with confidence 75%

{NOT beer} → {hot dogs} with confidence 11%



{beer} → {hot dogs} with confidence 31%

{NOT beer} → {hot dogs} with confidence 73%



You use the transaction data and compute the following tables

		hot dogs	NOT hot dogs	Total
single	beer	30	10	40
	NOT beer	5	40	45
NOT single	beer	40	88	128
	NOT beer	40	15	55

The left table allows to extract the following pairs of rules

$\{\text{beer}\} \rightarrow \{\text{hot dogs}\}$ with confidence 75%

$\{\text{NOT beer}\} \rightarrow \{\text{hot dogs}\}$ with confidence 11%



$\{\text{beer}\} \rightarrow \{\text{hot dogs}\}$ with confidence 31% (40/128)

$\{\text{NOT beer}\} \rightarrow \{\text{hot dogs}\}$ with confidence 73% (40/55)

{beer} → **{hot dogs}**

with confidence **42%**

{NOT beer} → **{hot dogs}**

with confidence **45%**



Customers who buy beer **{beer}** are less likely (**42%**) to buy hot dogs **{hot dogs}** than those who do not buy beer **{NOT beer}** (**45%**)



{beer} → **{hot dogs}**

with confidence **75%**

{NOT beer} → **{hot dogs}**

with confidence **11%**



Singles who buy beer **{beer}** are more likely (**75%**) to buy hot dogs **{hot dogs}** than those who do not buy beer **{NOT beer}** (**11%**)



Customers who buy beer
than those who do not



dogs {hot dogs}



Singles who buy beer {beer} are more likely (75%) to buy hot dogs {hot dogs} than those who do not buy beer {NOT beer} (11%)